KAIZEN MATHS

# Lesson starter

**Factorising Quadratic Expressions**

Complete Factorisation

Name _____

**S1**

Factorise each expression.

Factorise: $2x^2 - 20x + 48$

**S2**

Factorise each expression.

Factorise: $-6x^2 - 12x + 90$

**S3**

Factorise each expression.

Factorise: $36x^2 - 16$

**S4**

Factorise each expression.

Factorise: $2x^2 + 26x + 84$



Show your working

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ALG-CFE2-L3 · S

Show your working

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# Exit ticket

**Factorising Quadratic Expressions**

Complete Factorisation

Name _____

**E1**

Factorise each expression.

Factorise: $-2x^2 - 6x + 20$

**E2**

Factorise each expression.

Factorise: $150x^2 - 24$

**E3**

Factorise each expression.

Factorise: $6x^2 - 6x - 36$



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ALG-CFE2-L3 · E

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ALG-CFE2-L3 · E

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Factorise: $-2x^2 - 6x + 20$

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Factorise each expression.

Factorise: $150x^2 - 24$

E3

Factorise each expression.

Factorise: $6x^2 - 6x - 36$

ALG-CFE2-L3 · E

Show your working

# Teacher answers & working

**Factorising Quadratic Expressions · Complete Factorisation**

ALG-CFE2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S1

Factorise each expression.

Factorise: $2x^2 - 20x + 48$

**Answer:**

$2\,(x - 6)\,(x - 4)$

**Working:**

Take out the highest common factor before factorising the quadratic.

$2x^2 - 20x + 48 = 2\left(x^2 - 10x + 24\right)$

Now factorise the quadratic inside the brackets.

$-6 \times -4 = 24 \quad \text{and} \quad -6 + -4 = -10$

$2\left(x^2 - 10x + 24\right) = 2\,(x - 6)\,(x - 4)$

The expression is now fully factorised.

## S2

Factorise each expression.

Factorise: $-6x^2 - 12x + 90$

**Answer:**

$-6\,(x + 5)\,(x - 3)$

**Working:**

Take out the highest common factor before factorising the quadratic.

$-6x^2 - 12x + 90 = -6\left(x^2 + 2x - 15\right)$

Now factorise the quadratic inside the brackets.

$5 \times -3 = -15 \quad \text{and} \quad 5 + -3 = 2$

$-6\left(x^2 + 2x - 15\right) = -6\,(x + 5)\,(x - 3)$

The expression is now fully factorised.

# Teacher answers & working

**Factorising Quadratic Expressions · Complete Factorisation**

ALG-CFE2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S3

Factorise each expression.

Factorise: $36x^2 - 16$

**Answer:**

$4\,(3x - 2)\,(3x + 2)$

**Working:**

Take out the common factor first.

$36x^2 - 16 = 4\left(9x^2 - 4\right)$

The bracket is now a difference of two squares.

$A^2 - B^2 = (A - B)\,(A + B)$

$A = 3x, \quad B = 2$

$4\left(9x^2 - 4\right) = 4\,(3x - 2)\,(3x + 2)$

## S4

Factorise each expression.

Factorise: $2x^2 + 26x + 84$

**Answer:**

$2\,(x + 6)\,(x + 7)$

**Working:**

Take out the highest common factor before factorising the quadratic.

$2x^2 + 26x + 84 = 2\left(x^2 + 13x + 42\right)$

Now factorise the quadratic inside the brackets.

$6 \times 7 = 42 \quad \text{and} \quad 6 + 7 = 13$

$2\left(x^2 + 13x + 42\right) = 2\,(x + 6)\,(x + 7)$

The expression is now fully factorised.

# Teacher answers & working

**Factorising Quadratic Expressions · Complete Factorisation**

ALG-CFE2-L3 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## E1

Factorise each expression.

Factorise: $-2x^2 - 6x + 20$

**Answer:**

$-2\,(x - 2)\,(x + 5)$

**Working:**

Take out the highest common factor before factorising the quadratic.

$-2x^2 - 6x + 20 = -2\left(x^2 + 3x - 10\right)$

Now factorise the quadratic inside the brackets.

$-2 \times 5 = -10 \quad \text{and} \quad -2 + 5 = 3$

$-2\left(x^2 + 3x - 10\right) = -2\,(x - 2)\,(x + 5)$

The expression is now fully factorised.

## E2

Factorise each expression.

Factorise: $150x^2 - 24$

**Answer:**

$6\,(5x - 2)\,(5x + 2)$

**Working:**

Take out the common factor first.

$150x^2 - 24 = 6\left(25x^2 - 4\right)$

The bracket is now a difference of two squares.

$A^2 - B^2 = (A - B)\,(A + B)$

$A = 5x, \quad B = 2$

$6\left(25x^2 - 4\right) = 6\,(5x - 2)\,(5x + 2)$

# Teacher answers & working

**Factorising Quadratic Expressions · Complete Factorisation**

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## E3

Factorise each expression.

Factorise: $6x^2 - 6x - 36$

**Answer:**

$6\,(x + 2)\,(x - 3)$

**Working:**

Take out the highest common factor before factorising the quadratic.

$6x^2 - 6x - 36 = 6\left(x^2 - x - 6\right)$

Now factorise the quadratic inside the brackets.

$2 \times -3 = -6 \quad \text{and} \quad 2 + -3 = -1$

$6\left(x^2 - x - 6\right) = 6\,(x + 2)\,(x - 3)$

The expression is now fully factorised.